Rarity of colon cancer in Africans is associated with low animal product consumption, not fiber.

OBJECTIVE: To investigate whether the rarity of colon cancer in black Africans (prevalence, < 1:100,000) can be accounted for by dietary factors considered to reduce risk, and by differences in colonic bacterial fermentation. METHODS: Samples of the adult black South African population were drawn from several rural and urban regions. Food consumption was assessed by home visits, food frequency questionnaires, computerized analysis of 72-h dietary recall, and blood sampling. Colonic fermentation was measured by breath H_2 and CH_4 response to a traditional meal, and to 10-g of lactulose. Cancer risk was estimated by measurement of epithelial proliferation indices (Ki-67 and BrdU) in rectal mucosal biopsies. Results were evaluated by comparison to measurements in high-risk white South Africans (prevalence, 17:100,000). RESULTS: Epithelial proliferation was significantly lower in rural and urban blacks than whites. The diets of all the black subgroups were characterized by a low animal product and high boiled maize-meal content, whereas whites consumed more fresh animal products, cheese, and wheat products. Blacks consumed below RDA quantities of fiber (43% of RDA), vitamin A (78%), C (62%), folic acid (80%) and calcium (67%), whereas whites consumed more animal protein (177% of RDA) and fat (153%). Fasting and food-induced breath methane production was two to three times higher in blacks. CONCLUSIONS: The low prevalence of colon cancer in black Africans cannot be explained by dietary "protective" factors, such as, fiber, calcium, vitamins A, C and folic acid, but may be influenced by the absence of "aggressive" factors, such as excess animal protein and fat, and by differences in colonic bacterial fermentation.